

1 Rules

Congratulations on qualifying for the Tiebreaker Round! You will have **30** minutes for these 4 problems. You may turn in this answer sheet anytime before the 30 minute mark, but you may only turn it in once. Ties on the Individual Round will be broken by 1) number of correct answers on Tiebreaker Round, and 2) time at which the Tiebreaker Round was submitted.

2 Problems

- Find the smallest positive integer N so that one can select 7 distinct positive divisors of N , no two of which divide each other.
- Two frogs, Vivian and Luke, start at $(0, 0)$ and jump to $(3, 3)$. Vivian can move either one unit right or one unit up on each move. Meanwhile, Luke can move one unit right, one unit up, or diagonally one unit right and up on each move. Each frog's path is drawn out on the grid. Compute the probability that all of the closed polygons formed by their paths do NOT form a total area of 1 or greater.
- In triangle $\triangle ABC$, point E is on line AC and F is on line AB , and points E' and F' are the reflections of E across the midpoint of AC and F across the midpoint of AB . Suppose that segment EF and line $E'F'$ intersect at point D on segment BC . If $\frac{DE}{DF} = \frac{3}{5}$ and $\frac{EE'}{FF'} = \frac{1}{3}$, what is $\frac{AE}{AF}$?
- Alice and Bob play a game. They start with a number $N = 2^p 3^q 5^r 7^s$ for nonnegative integers p, q, r, s , and take turns, with Alice going first, dividing the number N by an integer from 2 through 9 such that the quotient is a positive integer. The player who divides the number such that the quotient is 1 wins the game. Compute the smallest three-digit number N for which Bob will win the game.

3 Answers

Problem 1. 	Problem 2.
Problem 3. 	Problem 4.